SUROJIT SAHA

surojit.gnit@gmail.com • +1-(385)-419-6427 • surojit-utah.github.io • LinkedIn: surojit-saha-6258ab15

EDUCATION

Ph.D. in Computer Science, ***Kahlert School of Computing, The University of Utah, USA*** 2018 – 2024
Research areas**:** Machine Learning and Computer Vision, ***Advisor*****:** Dr. Ross Whitaker **3.98/4.0**
Dissertation Title: Deep Representation Learning for Statistical Analysis

Master of Technology in Computer Science, ***Homi Bhabha National Institute, India*** 2009 – 2012
Research area**:** Autonomous Navigation of Mobile Robots, ***Advisor*****:** Dr. Prabir Kumar Pal **80.3/100**

Bachelor of Technology in Computer Science, ***Maulana Abul Kalam Azad University of Technology, India*** 2005 – 2009
Research area**:** Database Management Systems, ***Advisor*****:** Dr. Arup Kumar Bhaumik **9.08/10**

Research Interests: Agentic AI, LLM finetuning, RL alignment, VAEs, GANs, Diffusion Models, Generative Models, Transformers, Few-Shot Learning, Self-Supervised Learning, and Multitask Learning, Probabilistic Forecasting.

PROFESSIONAL EXPERIENCE

Applied Scientist - II, **Microsoft Corporation, USA** Feb 2026 – Present
Building Agentic AI solutions for solving real-world complex problems in the domain of supply chain optimization.

Data Scientist, **Microsoft Corporation, USA** Nov 2024 – Feb 2026
Stochastic Optimization, Bayesian Statistics for data driven inferences in long-term forecasting models.

Graduate Research Assistant, **The University of Utah, USA**, ***Advisor*****:** Dr. Ross Whitaker Aug 2021 – Oct 2024
LLM finetuning, Representation Learning, Few-Shot Learning, Multitask Learning, and Gaussian Processes.

CV/NLP PhD Intern, **Ancestry.com, USA***, **Mentor***: Dr. Jack Reese May 2021 – Aug 2021
Developed regularization techniques for detecting paragraphs, tables, and other data layouts related to the *Family History Book*.

Scientific Officer, **Department of Atomic Energy, Government of India** Sept 2009 – Aug 2018
Developed accurate, robust, and efficient localization algorithms for indoor mobile robots based on the principle of Monte Carlo.

PROJECTS

Eval-Driven Development of Enterprise LLM Agents

- Built and evaluated a supply communication agent across frontier models and orchestrators (Copilot Studio, Cowork).
- Improved the TSR from 73% to 85% with the use of skills served through an MCP server.
- Reduced tool calls by 32% without compromising accuracy by moving skills to orchestrator-native delivery.

Statistical Scenario Synthesis for Agent Evaluation

- Designed a non-LLM eval scenario synthesis pipeline using samples from configurable probability distributions.
- Scenarios are fully reproducible, while varying seeds and distributions produces diverse, controlled test conditions.

Stochastic Optimization & Bayesian Statistics

- Uncertainty quantification using Bayesian statistics and development of LP-based constrained optimization models.
- A *novel* approach to fuse demand and supply distributions for improved buffer optimization (patent submitted).

DBayes-LoRA: Bayesian LoRA Fine-Tuning of LLMs with Automatic Rank Pruning (in preparation)

- Places a variational posterior over LoRA's latent bottleneck, with an automatic relevance determination (ARD) prior.
- Prunes the uninformative latent dimensions during training using the relevance score. On LLaMA-2-7B, this reduces a rank-32 adapter to a median of ~10 active dimensions per projection while producing comparable metric scores.

Generative Encoding Networks (*GEN*)

- Uses the *kernel density estimation* (KDE) to match the aggregate posterior distribution.
- Demonstrates the use of the learned latent representations for *outlier detection*.

Aggregate Variational Autoencoders (*AVAE*)

- Matches the aggregate posterior to alleviate the *posterior collapse* and *holes* encountered in the VAE.
- Uses KDE in high-dimensional latent spaces (dimensions > 100).

Automatic Relevancy Detection in the Variational Autoencoder (*ARD-VAE*)

- Discovers *relevant* axes in the VAE using a *hierarchical* prior (using Bayesian statistics).

- Demonstrates the effectiveness in *modeling data distributions* and finding *latent factors explaining the data*.

Few-Shot Segmentation (FSS) Methods Using Gaussian Processes (a Bayesian modeling framework**)**

- Developed an FSS method for interpreting microscopy images, named the ***GP-UNet***.
- Proposed an FSS method that can adapt to the *varying number of classes*, dubbed the ***AdaSemSeg***.

3D Computer Vision

- Proposed an alternative to pairwise *3D scan registration* for joint estimation of the camera poses.
- Designed computationally efficient solvers for joint estimation of the camera poses in small loops.

PUBLICATIONS

1. Surojit Saha, Sarang Joshi, and Ross Whitaker**,** *ARD-VAE: A Statistical Formulation to Find the Relevant Latent Dimensions of Variational Autoencoders*, Winter Conference on Applications of Computer Vision (WACV), 2025 (**oral presentation**).

2. Surojit Saha, Sarang Joshi, and Ross Whitaker**,** *Disentanglement analysis in deep latent variable models matching aggregate posterior distributions*, International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2025.

3. Xiwen Li, Rehman Mohammed, Tristalee Mangin**, Surojit Saha,** Kerry Kelly, and Ross Whitaker, Tolga Tasdizen**,** *Joint Audio-Visual Idling Vehicle Detection with Streamlined Input Dependencies*, IEEE/CVF Winter Conference on Applications of Computer Vision Workshop (WACVW), 2025.

4. Surojit Saha, and Ross Whitaker**,** *AdaSemSeg: An Adaptive Few-shot Semantic Segmentation of Seismic Facies*, IEEE Transactions on Geoscience and Remote Sensing, 2025.

5. Surojit Saha, Sarang Joshi, and Ross Whitaker**,** *Matching Aggregate Posteriors in the Variational Autoencoder*, International Conference on Pattern Recognition (ICPR), 2024 (**oral presentation**).

6. Surojit Saha*, Wasim Gazi*, Rehman Mohammed, Thomas Rapstine, Hayden Powers, and Ross Whitaker, *Multi-task Training as Regularization Strategy for Seismic Image Segmentation*, IEEE Geoscience and Remote Sensing Letters, 2023.

7. Xiwen Li, Tristalee Mangin, **Surojit Saha**, Rehman Mohammed, Evan Blanchard, Dillon Tang, Henry Poppe, Nathan Searle, Ouk Choi, Kerry Kelly, and Ross Whitaker, *Real-Time Idling Vehicles Detection Using Combined Audio-Visual Deep Learning,* International Conference on Intelligent Traffic and Transportation, 2023.

8. Surojit Saha, Shireen Elhabian, and Ross Whitaker**,** *GENs: Generative Encoding Networks*, Machine Learning, 2022.

9. Surojit Saha, Ouk Choi, and Ross Whitaker, *Few-Shot Segmentation of Microscopy Images Using Gaussian Process*, a MICCAI workshop on Medical Optical Imaging and Virtual Microscopy Image Analysis, 2022 (**best paper award**).

10. Pedro Miraldo, **Surojit Saha,** and Srikumar Ramalingam, *Minimal Solvers for Mini-Loop Closures in 3D Multi-Scan Alignment*, IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019.

11. Biswajit Sarkar, **Surojit Saha**, and Prabir K Pal, *A novel method for computation of importance weights in Monte Carlo localization online segment-based maps*, Robotics and Autonomous Systems, 2015.

12. Surojit Saha, Biswajit Sarkar, Prabir K Pal, *Monte Carlo-based pose tracking on maps represented with line segments*, Conference on Advances in Robotics, 2015.

ACHIEVEMENTS

1. Best paper award at MICCAI workshop on Medical Optical Imaging and Virtual Microscopy Image Analysis, 2022.
2. A part of my PhD was supported by the National Institutes of Health under award number R01ES032810.
3. 99.16 percentile in GATE-2009, India (countrywide exam for pursuing Master of Technology).

SERVICES

Conferences: Served as a reviewer for the ICASSP, WACV, AAAI, ACML, and ICVGIP.
Journals: Served as a reviewer for the IEEE TPAMI and IEEE TGRS, Scientific Reports.

TEACHING

1. *Teaching Mentor* for **Deep Learning**, Fall 2019, *Kahlert School of Computing, The University of Utah, USA*.
2. *Teaching Mentor* for **3D Computer Vision**, Spring 2020, *Kahlert School of Computing, The University of Utah, USA*.
3. *Delivered a tutorial on PyTorch* in **Image Processing**, Fall 2023, *The University of Utah, USA*.
4. *Delivered a lecture on Introduction to Statistics* in **Probability and Statistics for Engineers**, Spring 2024, *Kahlert School of Computing, The University of Utah, USA*.